

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyse DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Concept Mapping (Medium)
Assessment Tool Weightage: 20%

QUESTIONS		
Q.No	Question	Bloom's Level
1	Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.	BL2 – Understand
2	Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.	BL2 – Understand
3	Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.	BL2 – Understand
4	Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.	BL2 – Understand
5	Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.	BL2 – Understand
6	Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.	BL2 – Understand

7	Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.	BL2 – Understand
8	Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.	BL2 – Understand
9	Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.	BL2 – Understand
10	Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.	BL1 – Remember

Code of Conduct:

I R, INDHU (192524332)__(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

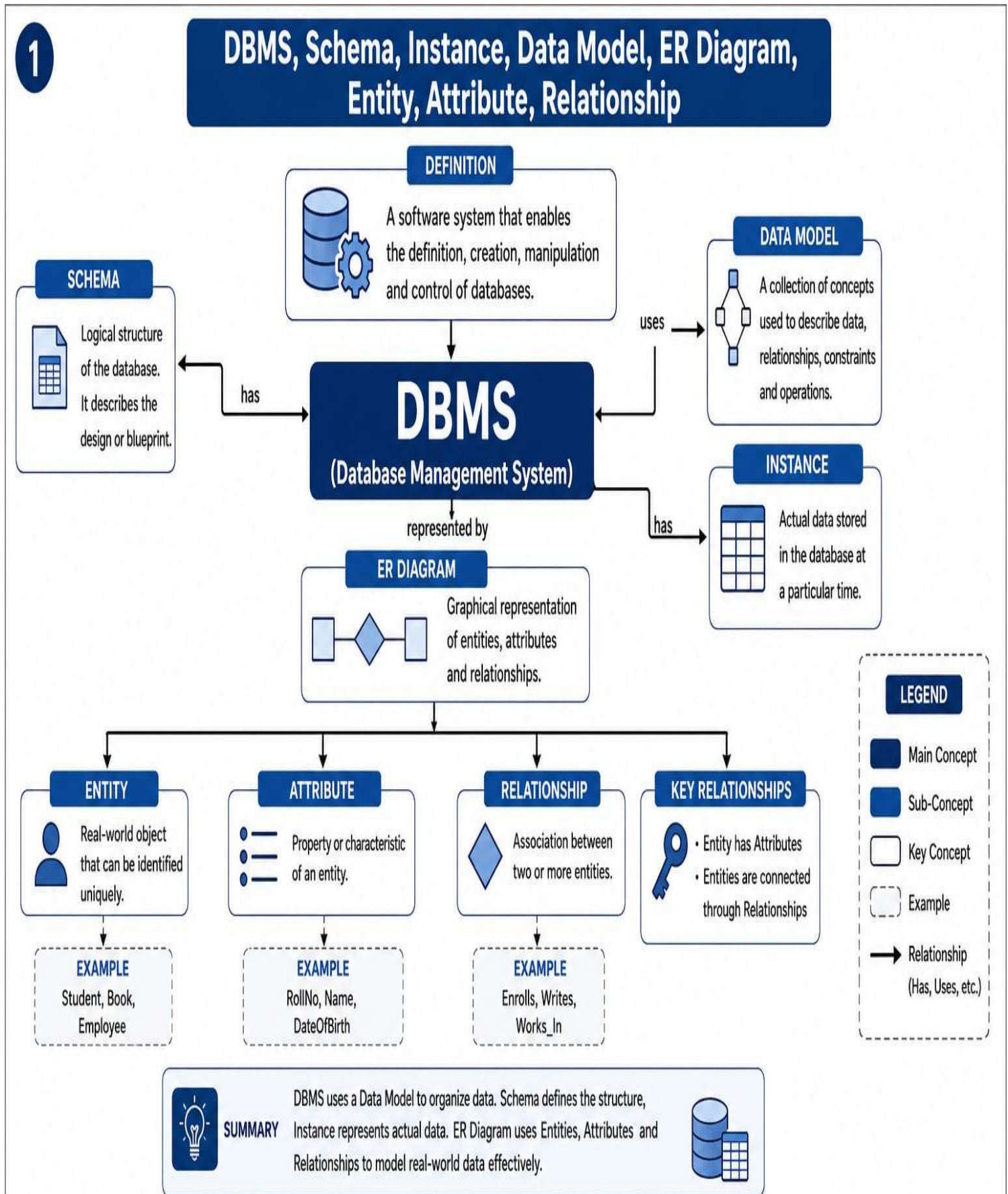
Signature of the student:
R. INDHU

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure

Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

1. Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.

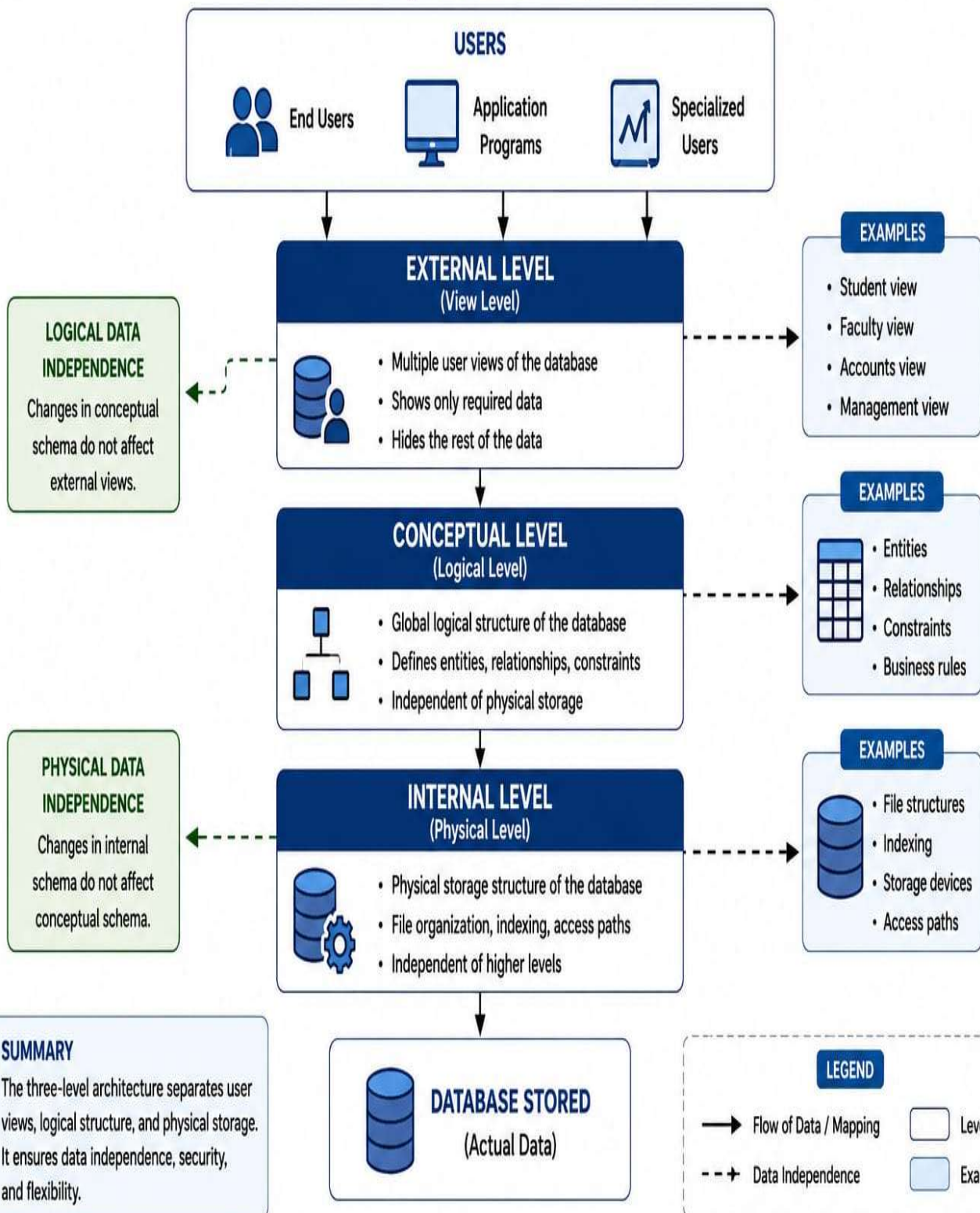


2. Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.

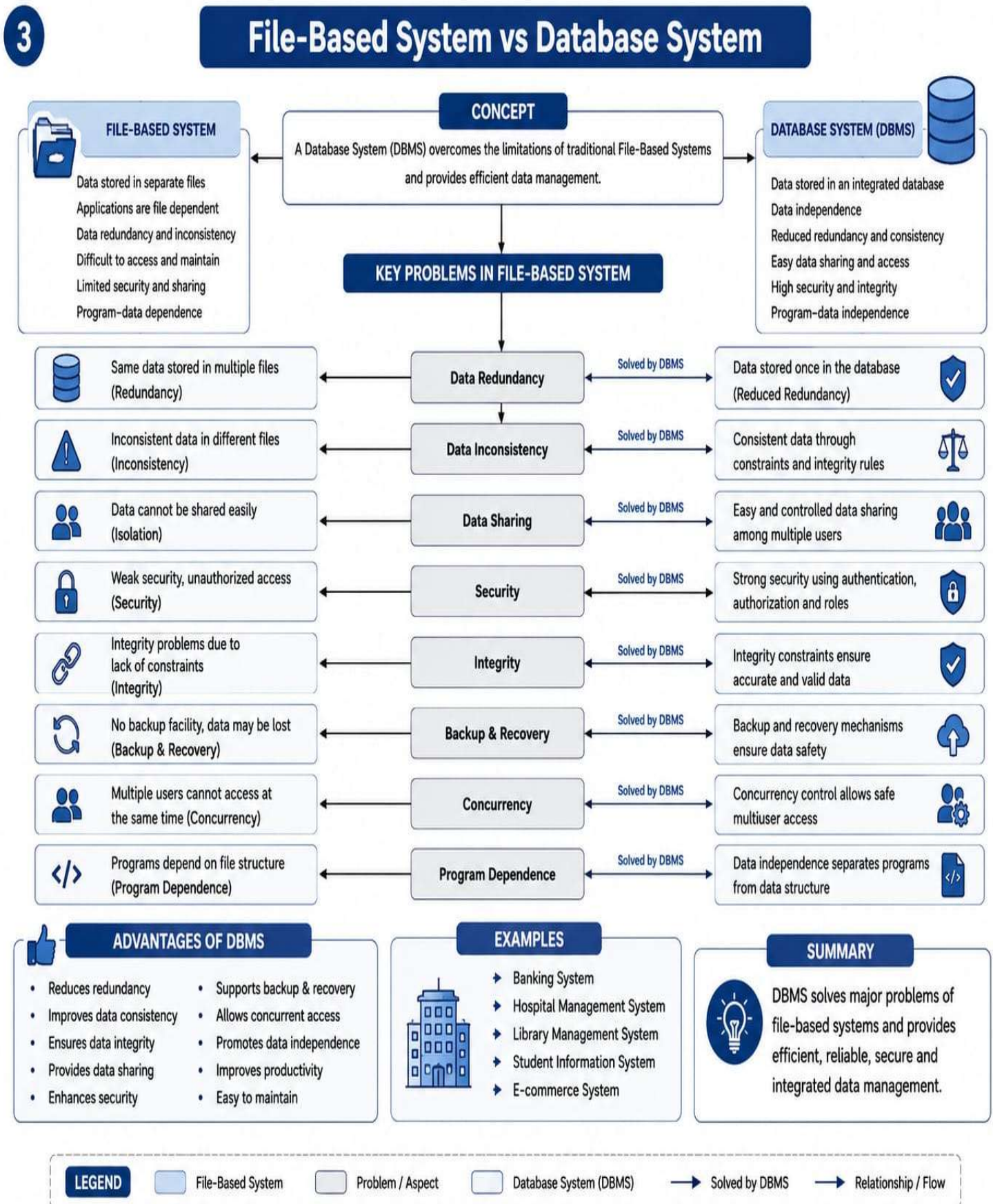
2

ANSI/SPARC THREE-LEVEL ARCHITECTURE

Provides Data Independence and Multiple User Views



3. Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.



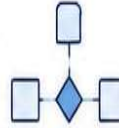
4. Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.

4

ER DIAGRAM COMPONENTS

DEFINITION

ER Diagram (Entity-Relationship Diagram) is a graphical representation of entities, attributes and relationships in a database.



ENTITIES

Strong Entity

An entity that has its own primary key.



Weak Entity

An entity that does not have a primary key and depends on another entity.



Entity Set

A collection of similar entities.



EXAMPLE



Student, Department, Employee

ATTRIBUTES

Simple Attribute

Cannot be divided.



Composite Attribute

Can be divided into sub-parts.



Multivalued Attribute

Can have multiple values for an entity.



Derived Attribute

Computed from other attributes.



Key Attribute

Uniquely identifies an entity.



EXAMPLE

Name (Simple),
Address (Composite),
PhoneNos (Multivalued),
Age (Derived),
RollNo (Key)



RELATIONSHIPS

Unary Relationship

Relationship between instances of the same entity.



Binary Relationship

Relationship between two entities.



Ternary Relationship

Relationship between three entities.



n-ary Relationship

Relationship between more than three entities.



EXAMPLE

Manages (Unary),
Works_In (Binary),
Supplies (Ternary)



PARTICIPATION CONSTRAINTS

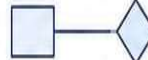
Total Participation

Every entity in the set must participate in the relationship.



Partial Participation

Some entities may or may not participate in the relationship.



EXAMPLE

Employee must work in a Department (Total) but Department may or may not have Employees (Partial)



CARDINALITY (MAPPING)



1:1 (One to One)

One entity in A is related to one entity in B.



1:N (One to Many)

One entity in A is related to many entities in B.



N:1 (Many to One)

Many entities in A are related to one entity in B.



M:N (Many to Many)

Many entities in A are related to many entities in B.

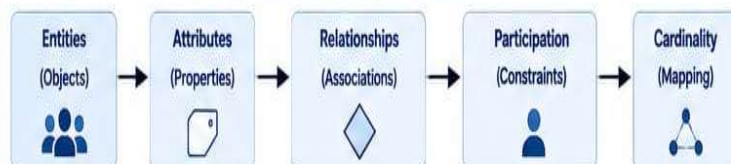


EXAMPLE

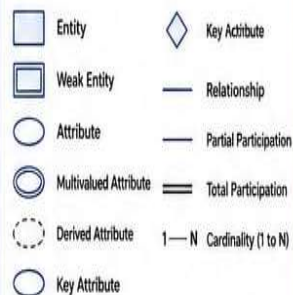
1:1 → Person-Passport
1:N → Department-Employee
N:1 → Employee-Department
M:N → Student-Course



SUMMARY OF COMPONENTS



LEGEND



KEY POINTS

- Entities represent real-world objects.
- Attributes describe the properties of entities.
- Relationships connect entities.
- Participation shows involvement of entities in relationships.
- Cardinality defines the number of occurrences related.

5. Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.

5

Hierarchical, Network and Relational Data Models

CONCEPT

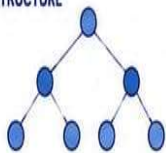


Data models are theoretical frameworks that define how data is organized, stored, related and manipulated in a database.



HIERARCHICAL MODEL

STRUCTURE



- Tree like structure
- One parent – many children
- Each child has only one parent
- Data flows from top to bottom

ADVANTAGES



- Simple and easy to understand
- Fast access to data
- Good performance for hierarchical data

DISADVANTAGES



- Inflexible structure
- Difficult to manage many to many relationships
- Not suitable for complex queries

APPLICATIONS



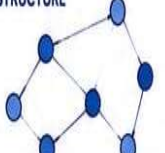
- Organization hierarchy
- File systems
- Company information systems

EXAMPLE

Company Organization
(Department – Employee)

NETWORK MODEL

STRUCTURE



- Graph structure
- Many – to – many relationships
- Each record may have multiple owners
- More complex than hierarchical model

ADVANTAGES



- Supports many-to-many relationships
- More flexible than hierarchical model
- Efficient for complex data

DISADVANTAGES



- Complex to design and navigate
- Higher maintenance cost
- Program-data dependence

APPLICATIONS



- Telecommunication networks
- Airline reservation systems
- Supply chain management

EXAMPLE

Supplier – Part – Project
(Many suppliers supply many parts
used in many projects)

RELATIONAL MODEL

STRUCTURE



- Data organized in tables (relations)
- Rows represent tuples (records)
- Columns represent attributes
- Relations linked using keys

ADVANTAGES



- Simple and easy to use
- Data independence
- Powerful query language (SQL)
- Easy to enforce integrity

DISADVANTAGES



- More storage overhead
- Complex queries may be slow
- Requires proper normalization

APPLICATIONS



- Banking systems
- ERP systems
- Student information systems

EXAMPLE

Student Table (RollNo, Name, Dept, Age)
Course Table (CourseID, CourseName)
Enrollment Table (RollNo, CourseID)

COMPARISON TABLE

FEATURE	HIERARCHICAL MODEL	NETWORK MODEL	RELATIONAL MODEL
Structure	Tree	Graph	Tables (Relations)
Relationship	One to Many	Many to Many	One to Many / Many to Many
Flexibility	Low	High	Very High
Complexity	Low	High	Low to Moderate
Data Independence	Low	Low	High
Query Language	Navigational	Navigational	Declarative (SQL)
Examples	File System, Org Chart	Telecom, Reservation	Banking, ERP, University

LEGEND

- Node / Record
- Relationship / Link
- Table / Relation
- 👍 Advantage
- 👎 Disadvantage
- ⚙️ Application
- Example

APPLICATIONS OVERVIEW



Banking



Hospital
Management



Library
System



E-Commerce



HR
Management



Telecommunication

SUMMARY



- Hierarchical model is simple but inflexible.
- Network model supports many-to-many but is complex.
- Relational model is most widely used due to its simplicity, flexibility and data independence.

6. Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation

6

ENHANCED ER (EER) MODEL

DEFINITION

The Enhanced Entity-Relationship (EER) Model extends the basic ER model by adding more concepts to represent real-world constraints and semantics.

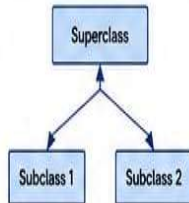


ER MODEL (BASIS)



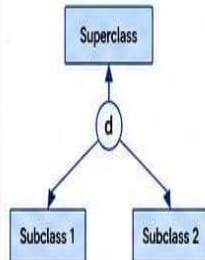
- Basic modeling using Entities, Attributes and Relationships.
- Does not capture higher level constraints.

GENERALIZATION



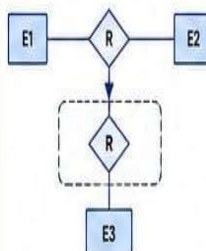
- Bottom-up approach.
- Combines multiple lower level entities into a higher level entity.

SPECIALIZATION



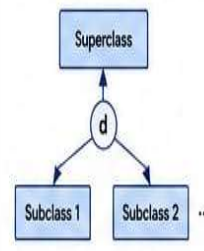
- Top-down approach.
- Splits a higher level entity into multiple lower level entities.

AGGREGATION



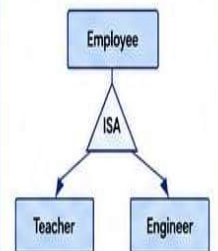
- Treats a relationship set as a higher level entity.
- Used to model relationships among relationships.

INHERITANCE



- Subclasses inherit attributes and relationships from superclass.
- Promotes reusability and reduces redundancy.

ISA RELATIONSHIP



- "IS A" relationship between superclass and subclass.
- Defines specialization or generalization.

KEY FEATURES OF EER MODEL



Captures Real-World Semantics



Supports Hierarchical Abstraction



Reduces Data Redundancy



Improves Data Integrity and Consistency

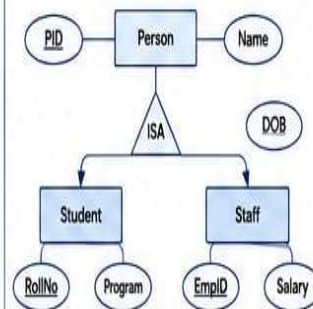


Flexible and Extensible



Better Design and Understandability

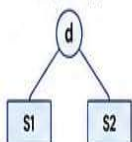
EXAMPLE (UNIVERSITY SYSTEM)



- Person is a superclass.
- Student and Staff are subclasses.
- Student has attributes RollNo, Program.
- Staff has attributes EmpID, Salary.
- ISA defines the relationship between superclass and subclasses.

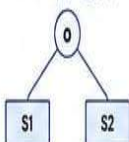
TYPES OF SPECIALIZATION

Disjoint (d)



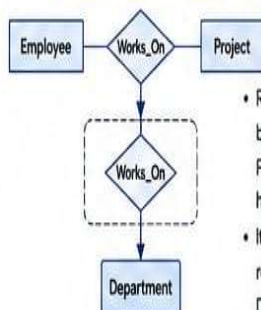
- An entity in superclass can belong to only one subclass.

Overlapping (o)



- An entity in superclass can belong to more than one subclass.

EXAMPLE OF AGGREGATION



- Relationship Works_On between Employee and Project is treated as a higher level entity.
- It participates in another relationship with Department.

ADVANTAGES

- ✓ Better modeling of real-world scenarios
- ✓ Supports complex business rules
- ✓ Reduces redundancy
- ✓ Improves integrity and consistency
- ✓ Flexible and scalable design

DISADVANTAGES

- ✗ Complex to design and understand
- ✗ Increases system complexity
- ✗ Requires more development time
- ✗ Not supported in all DBMS
- ✗ Higher maintenance cost

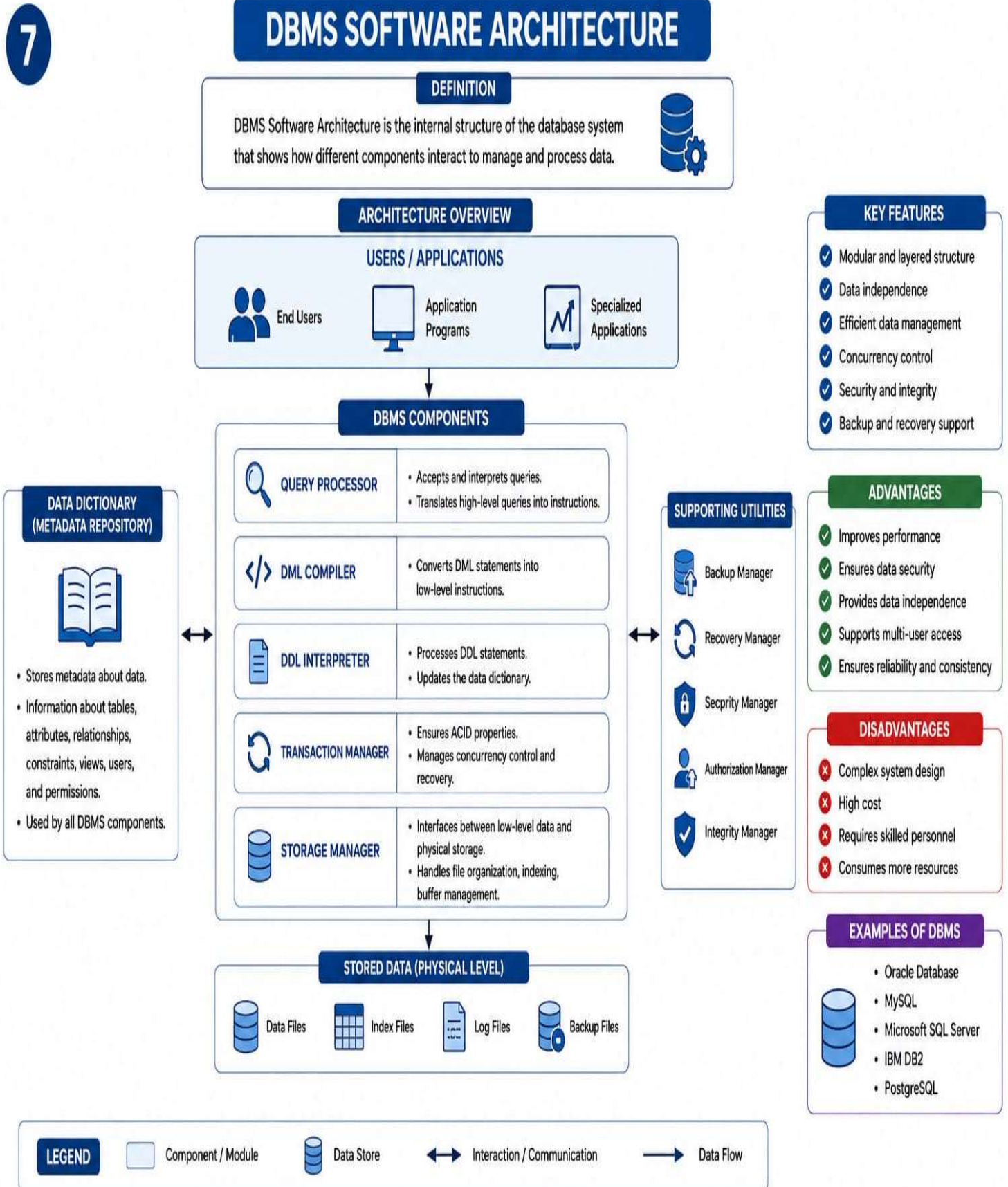
LEGEND



SUMMARY

The EER Model extends the ER Model by adding Generalization, Specialization, Aggregation, Inheritance and ISA relationships. It provides powerful tools to design databases that closely represent real-world requirements.

7. Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.



8. Map the data models discussed in CO1 to real-world applications and explain the suitability of each model

8

NORMALIZATION (1NF, 2NF, 3NF)

CONCEPT

Normalization is the process of organizing data in a database to minimize redundancy and improve data integrity by dividing large tables into smaller, well-structured tables.



NORMALIZATION PROCESS

UNNORMALIZED FORM (UNF)

- Repeating groups exist
- Redundant data
- Inconsistent results



FIRST NORMAL FORM (1NF)

- Eliminate repeating groups
- Each field contains atomic (single) values
- Unique rows



SECOND NORMAL FORM (2NF)

- In 1NF
- Remove partial dependencies
- Every non-key attribute is fully dependent on the primary key



THIRD NORMAL FORM (3NF)

- In 2NF
- Remove transitive dependencies
- Non-key attributes depend only on the primary key, not on other non-key attributes



BCNF (Optional)

- Stronger than 3NF
- For every functional dependency $X \rightarrow Y$, X should be a super key



NORMAL FORMS SUMMARY

- 1NF** – Atomic values, no repeating groups
- 2NF** – 1NF + No partial dependency
- 3NF** – 2NF + No transitive dependency
- BCNF** – Every determinant is a candidate key

EXAMPLE: STUDENT ENROLLMENT

UNF

Student_Info (Unnormalized)

SID	SName	Courses	Instructor
1	Ramesh	Math, DBMS	Dr.A, Dr.B
2	Sita	OS, CN	Dr.C, Dr.D
3	Kumar	DBMS, OS	Dr.B, Dr.C

1NF

Student_Info (1NF)

SID	SName	Course	Instructor
1	Ramesh	Math	Dr.A
1	Ramesh	DBMS	Dr.B
2	Sita	OS	Dr.C
2	Sita	CN	Dr.D
3	Kumar	DBMS	Dr.B
3	Kumar	OS	Dr.C

2NF

Student (2NF)

SID	SName
1	Ramesh
2	Sita
3	Kumar

Enrollment (2NF)

SID	Course
1	Math
1	DBMS
2	OS
2	CN
3	DBMS
3	OS

Course_Instructor (2NF)

Course	Instructor
Math	Dr.A
DBMS	Dr.B
OS	Dr.C
CN	Dr.D

3NF

Student (2NF)

SID	SName
1	Ramesh
2	Sita
3	Kumar

Enrollment (2NF)

SID	Course
1	Math
1	DBMS
2	OS
2	CN
3	DBMS
3	OS

Course (3NF)

Course	CourseName
Math	Mathematics
DBMS	Database Mgmt
OS	Operating System
CN	Computer Networks

Instructor (3NF)

InstructorID	InstructorName
Dr.A	Doctor A
Dr.B	Doctor B
Dr.C	Doctor C
Dr.D	Doctor D

Course_Instructor (3NF)

Course	InstructorID
Math	Dr.A
DBMS	Dr.B
OS	Dr.C
CN	Dr.D

BENEFITS OF NORMALIZATION

- Reduce data redundancy
- Improve data integrity and consistency
- Avoid insertion, update and deletion anomalies
- Save storage space
- Easy to maintain and modify data

ANOMALIES SOLVED

- Insertion Anomaly
- Update Anomaly
- Deletion Anomaly

GENERAL RULES OF NORMALIZATION

- Each table should have a primary key.
- Non-key attributes should depend on the key, the whole key and nothing but the key.
- Each field should contain atomic (single) values.
- Eliminate redundant and unnecessary data.

KEY POINTS

- Normalization organizes data into related tables.
- It improves database design and performance.
- 1NF, 2NF and 3NF are the most commonly used forms.
- Higher normal forms lead to better data quality.

LEGEND

- Table
- Intermediate Result
- Transformation / Decomposition
- Relationship (Key Reference)

REAL LIFE EXAMPLE APPLICATIONS



Banking Systems



E-Commerce Systems



Student Information Systems



Hospital Management Systems

9. Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence

9

TRANSACTIONS IN DBMS (ACID PROPERTIES)

CONCEPT

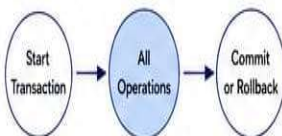
A Transaction is a sequence of database operations treated as a single logical unit of work. It must be reliable and maintain the consistency of the database.



ACID PROPERTIES

A - ATOMICITY

The transaction is "all or nothing". Either all operations are completed or none are.

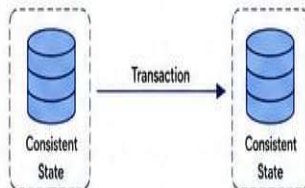


Example

If money is transferred from A to B, either both debit and credit occur or none occur.

C - CONSISTENCY

A transaction brings the database from one consistent state to another. It maintains all integrity constraints.

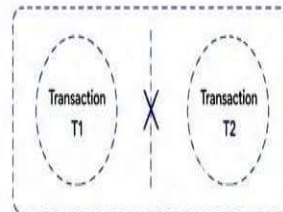


Example

The sum of account balances before and after the transaction remains the same.

I - ISOLATION

Concurrent transactions do not interfere with each other. Each transaction is isolated until it is complete.

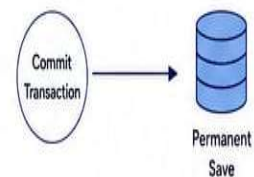


Example

While T1 is executing, T2 cannot see the intermediate results of T1.

D - DURABILITY

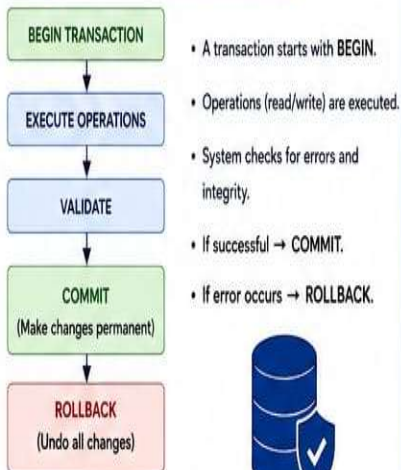
Once a transaction is committed, its changes are permanent even if a system failure occurs.



Example

After commit, the data remains in the database even after a crash.

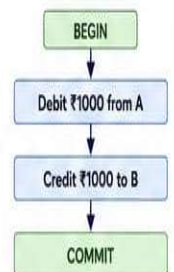
TRANSACTION PROCESS



EXAMPLE: BANK TRANSFER

Successful Transaction

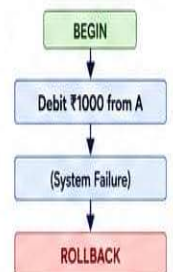
Transfer ₹1000 from A to B



Result:
Both operations completed.
Database is consistent.

Failed Transaction

Transfer ₹1000 from A to B

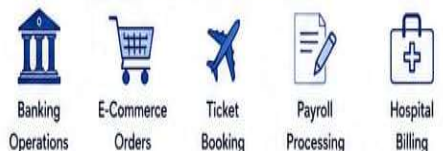


Result:
Transaction rolled back.
No changes made.

IMPORTANCE

- ✓ Ensures data reliability and consistency.
- ✓ Prevents data corruption.
- ✓ Supports concurrent access safely.
- ✓ Essential for real-time and multi-user systems.

EXAMPLES OF TRANSACTIONS



KEY POINTS

- ✓ Transactions follow ACID properties.
- ✓ They maintain data integrity and reliability.
- ✓ Commit makes changes permanent.
- ✓ Rollback undoes all changes.

LEGEND

- Start / End
- Process / Operation
- Rollback / Failure
- Database
- Isolation Boundary

10. Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application D

10

Concurrency Control and Recovery in DBMS

CONCEPT

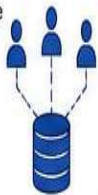
Concurrency control allows multiple transactions to execute simultaneously without interfering with each other.
Recovery ensures the database is restored to a correct state after a failure.



CONCURRENCY CONTROL

NEED

- Many users access the database at the same time.
- Prevents conflicts between simultaneous transactions.
- Maintains data consistency and integrity.



PROBLEMS WITHOUT CONCURRENCY CONTROL

- Lost Update Problem
- Dirty Read (Uncommitted Dependency)
- Non-Repeatable Read
- Phantom Read



CONCURRENCY CONTROL TECHNIQUES

1. LOCK BASED PROTOCOLS



- Data items are locked before access.
- Other transactions must wait.
- Types of Locks:
 - Shared Lock (Read)
 - Exclusive Lock (Write)

2. TIMESTAMP BASED PROTOCOLS



- Each transaction gets a unique timestamp.
- Operations are executed based on timestamps.
- Ensures serializability.

SCHEDULES

SERIAL SCHEDULE

Transactions execute one after another.



Always produces correct result.

CONCURRENT SCHEDULE

Transactions execute in an interleaved way.



Needs concurrency control to ensure correctness.

RECOVERY

NEED

- System failures may cause loss of data.
- Recovery restores the database to a consistent state.
- Ensures durability and reliability.

TYPES OF FAILURES

- Transaction Failure – logical errors, deadlock, etc.
- System Failure – power failure, crash, etc.
- Media Failure – disk crash, corruption, etc.

RECOVERY TECHNIQUES

1. LOG BASED RECOVERY



- All updates are recorded in a log.
- Uses Write-Ahead Logging (WAL).
- Types:
 - Undo: Rollback uncommitted transactions.
 - Redo: Reapply committed transactions.

2. CHECKPOINTING



- System saves a checkpoint periodically.
- Reduces the amount of log needed during recovery.

RECOVERY PROCESS



COMPARISON

ASPECT	CONCURRENCY CONTROL	RECOVERY
Purpose	Manages simultaneous access by multiple transactions	Restores database after failures
Focus	Prevents conflicts	Restores correct state
Problems Solved	Lost update, dirty read, non-repeatable read, phantom read	Data loss, incomplete transactions
Techniques	Lock based, Timestamp based	Log based recovery, Checkpointing
Ensures	Consistency & Isolation	Durability & Reliability

IMPORTANCE



Ensures data accuracy and consistency



Allows multiple users to work efficiently



Protects data from failures and crashes



Essential for real-time and large-scale systems.

EXAMPLES



Banking Systems



E-Commerce Systems



Airline Reservation



Hospital Management



University Systems

LEGEND:



Outcome / Result



Technique / Method



Process / Step



User



Database



Log / Record



Warning



Checkpoint